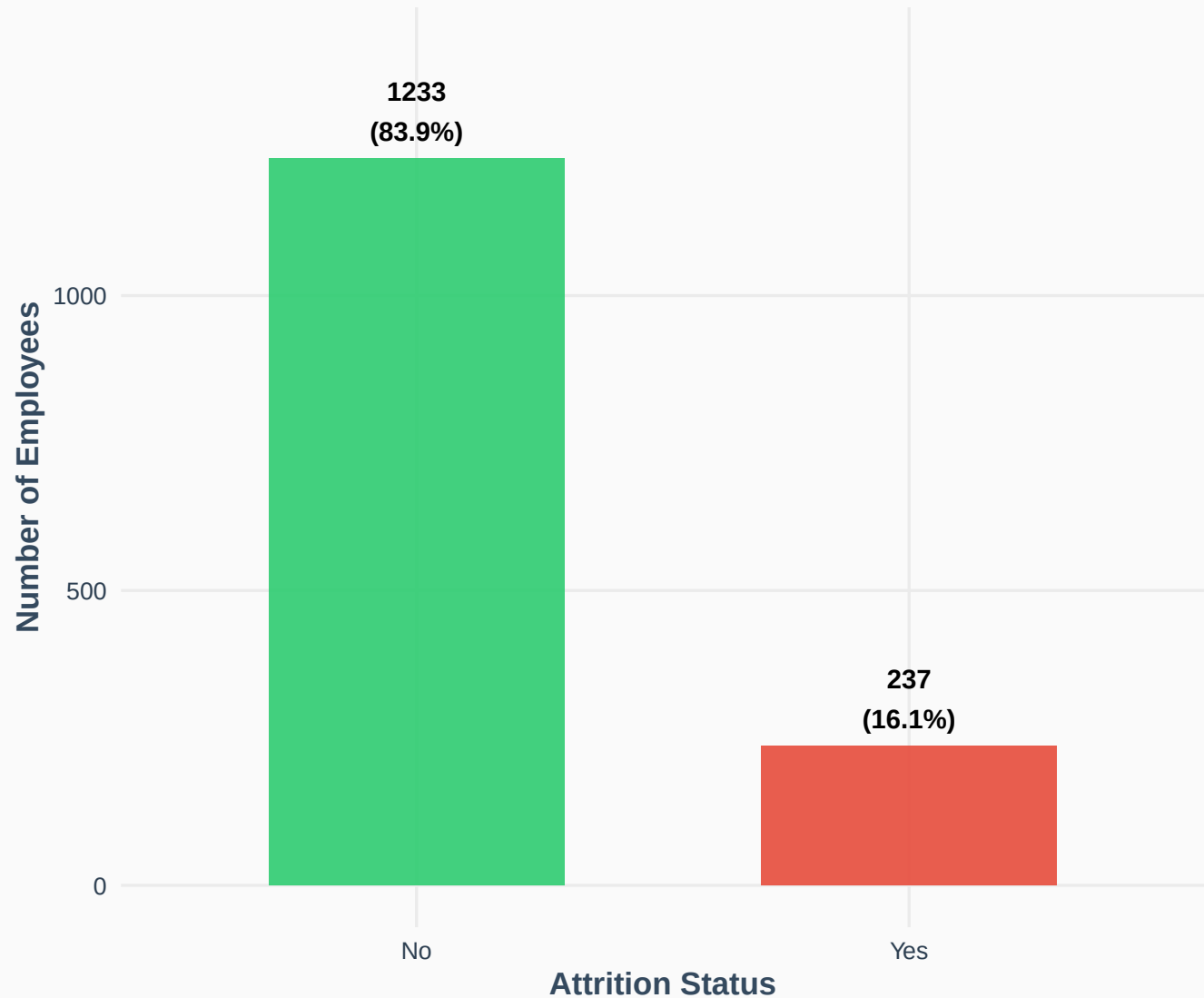


Employee Attrition Distribution

Imbalanced dataset — ~16% attrition rate

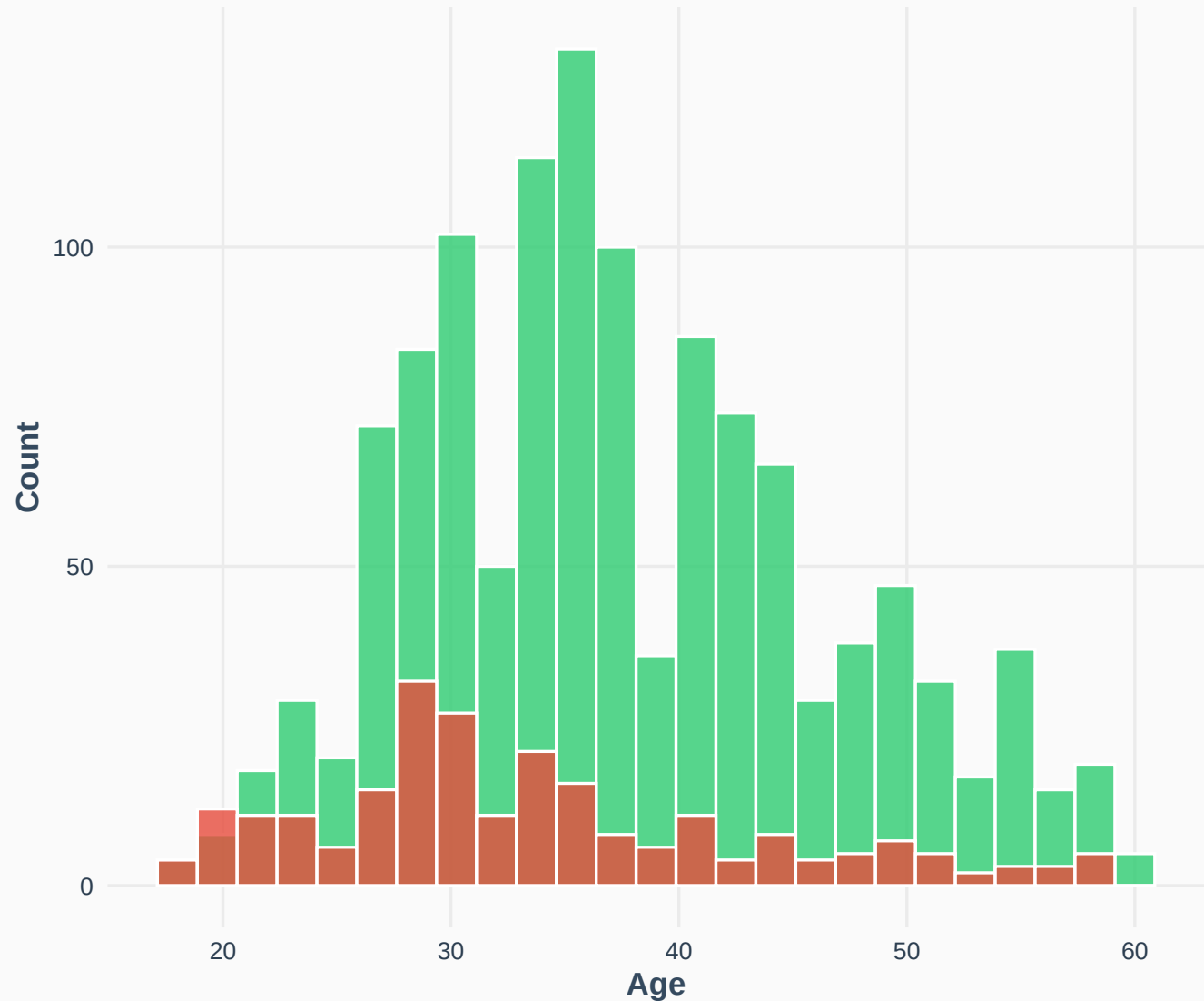
Attrition ■ No ■ Yes



Age Distribution by Attrition

Younger employees tend to have higher attrition rates

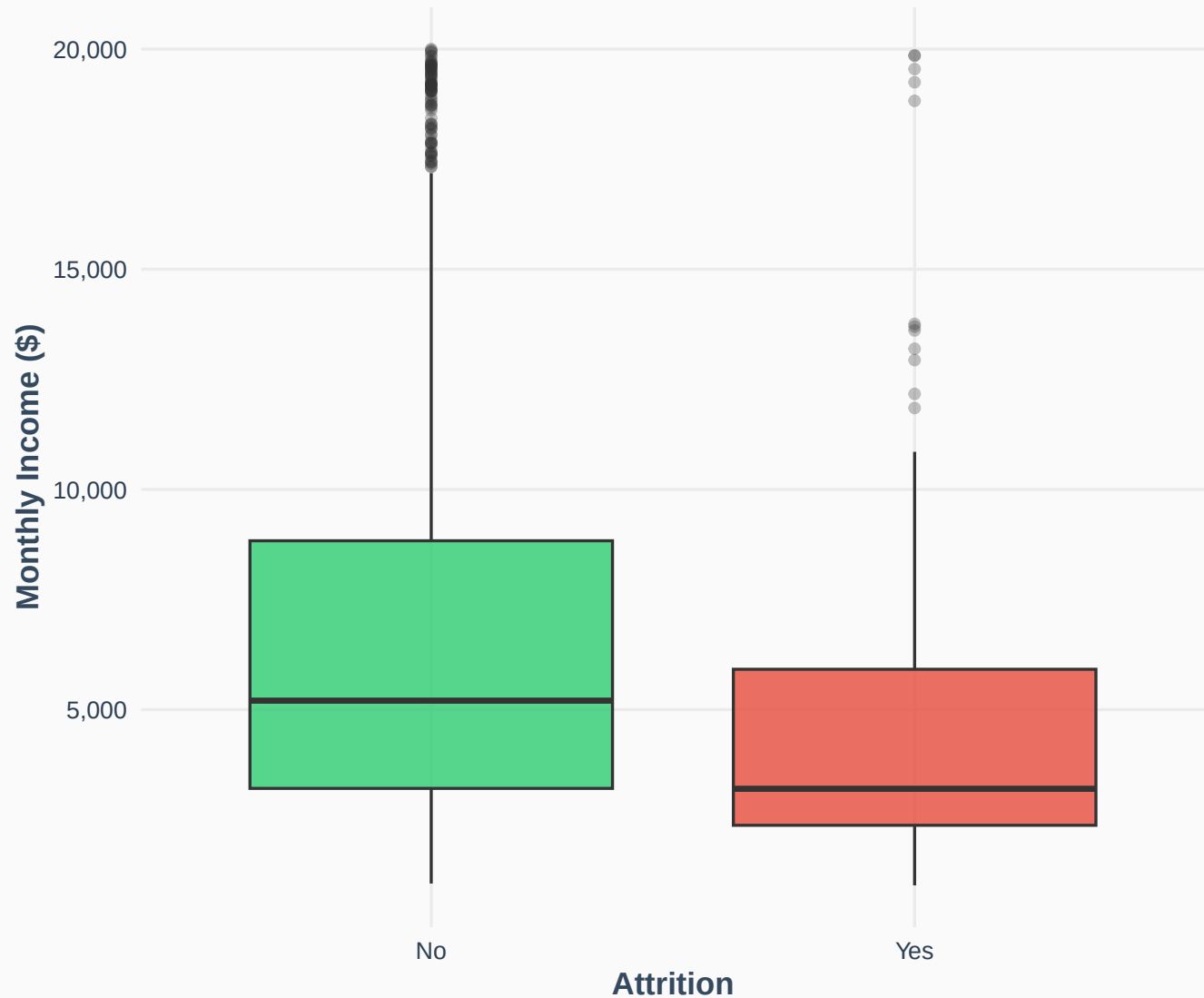
Attrition ■ No ■ Yes



Monthly Income by Attrition Status

Employees who leave tend to earn less

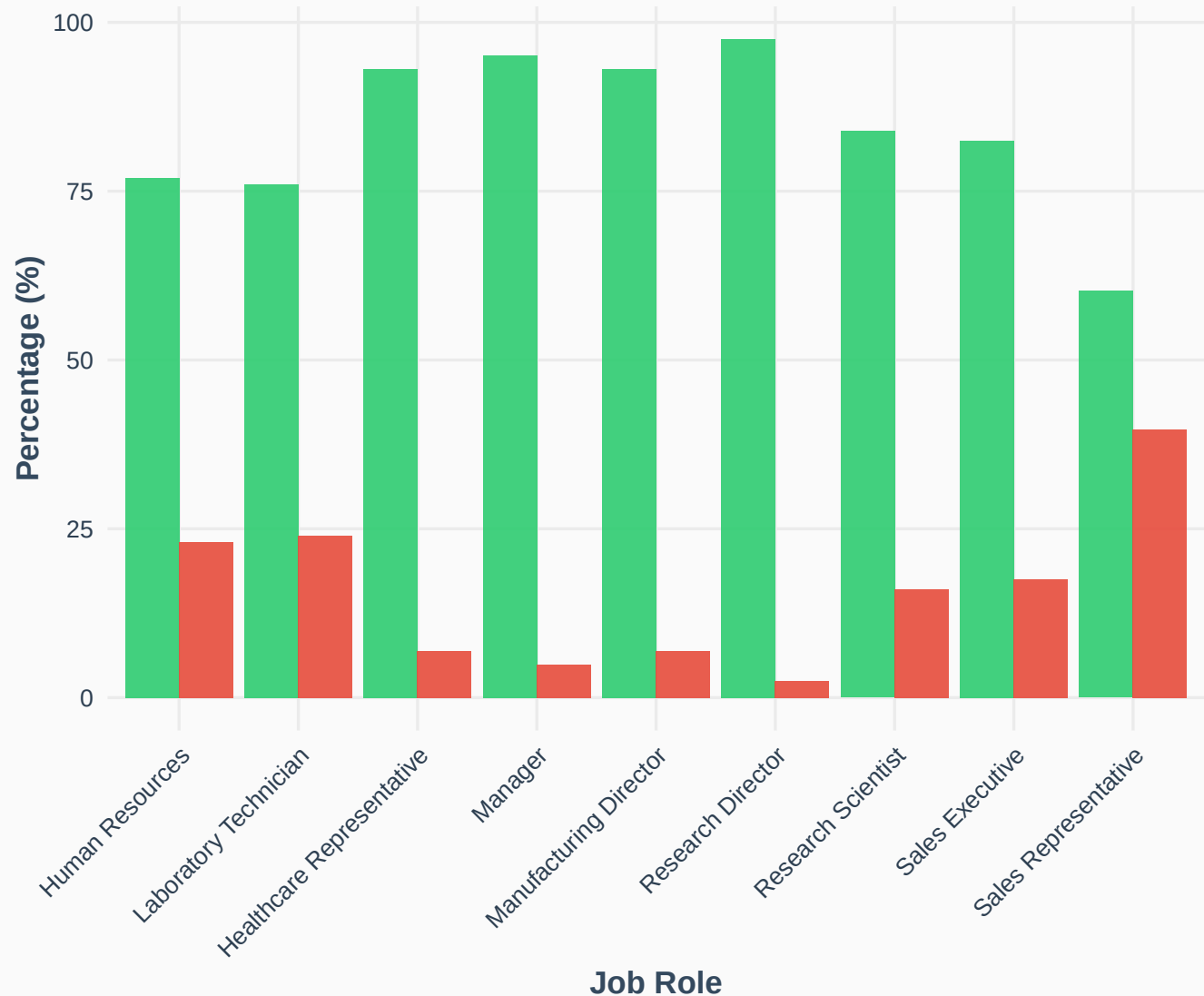
Attrition  No  Yes



Attrition Rate by Job Role

Some roles experience significantly higher turnover

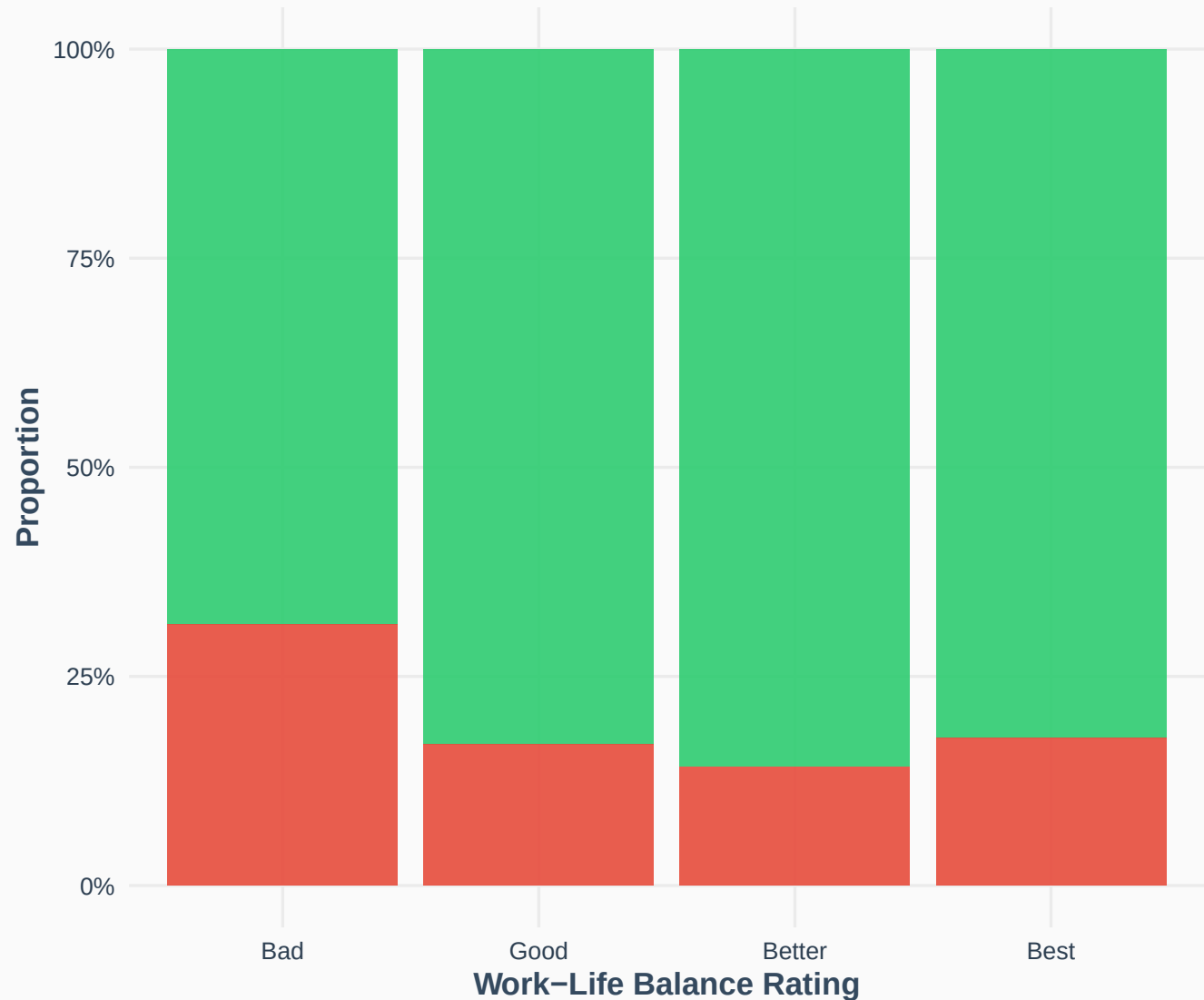
Attrition ■ No ■ Yes



Attrition Rate by Work-Life Balance

Poor work-life balance correlates with higher attrition

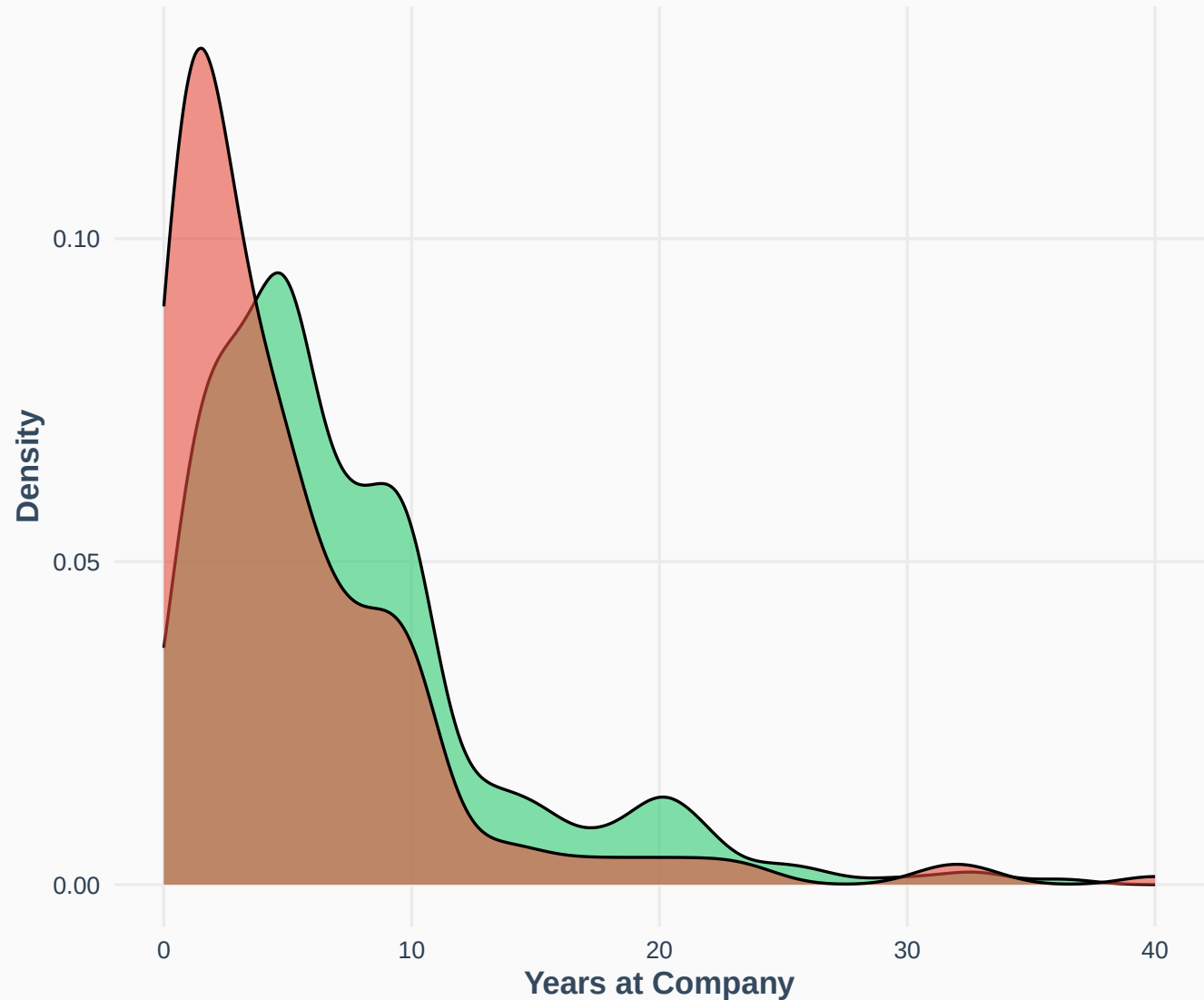
Attrition ■ No ■ Yes



Years at Company — Density by Attrition

Employees with fewer years at the company are more likely to leave

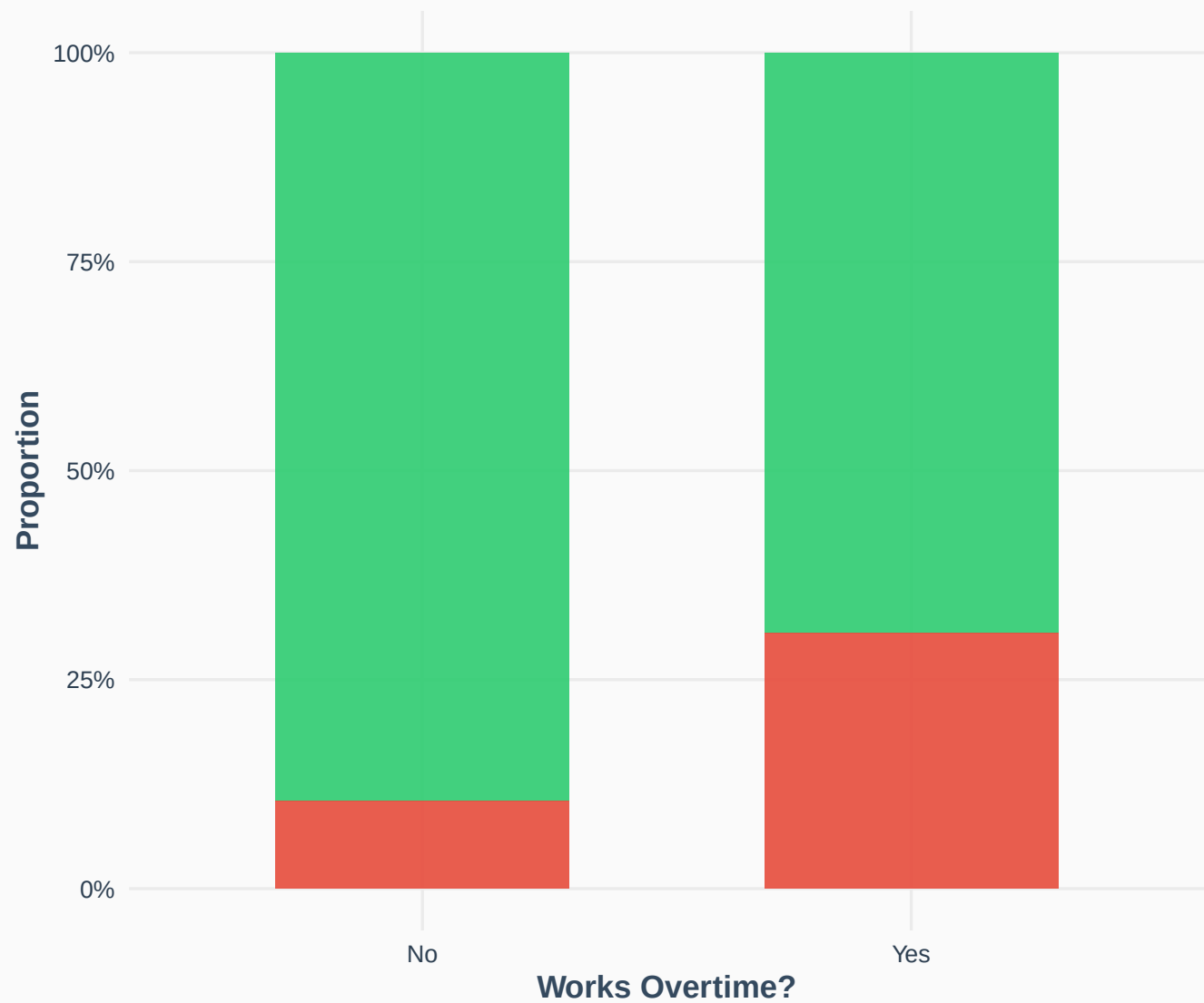
Attrition ■ No ■ Yes



Attrition Rate by Overtime Status

Employees working overtime are significantly more likely to leave

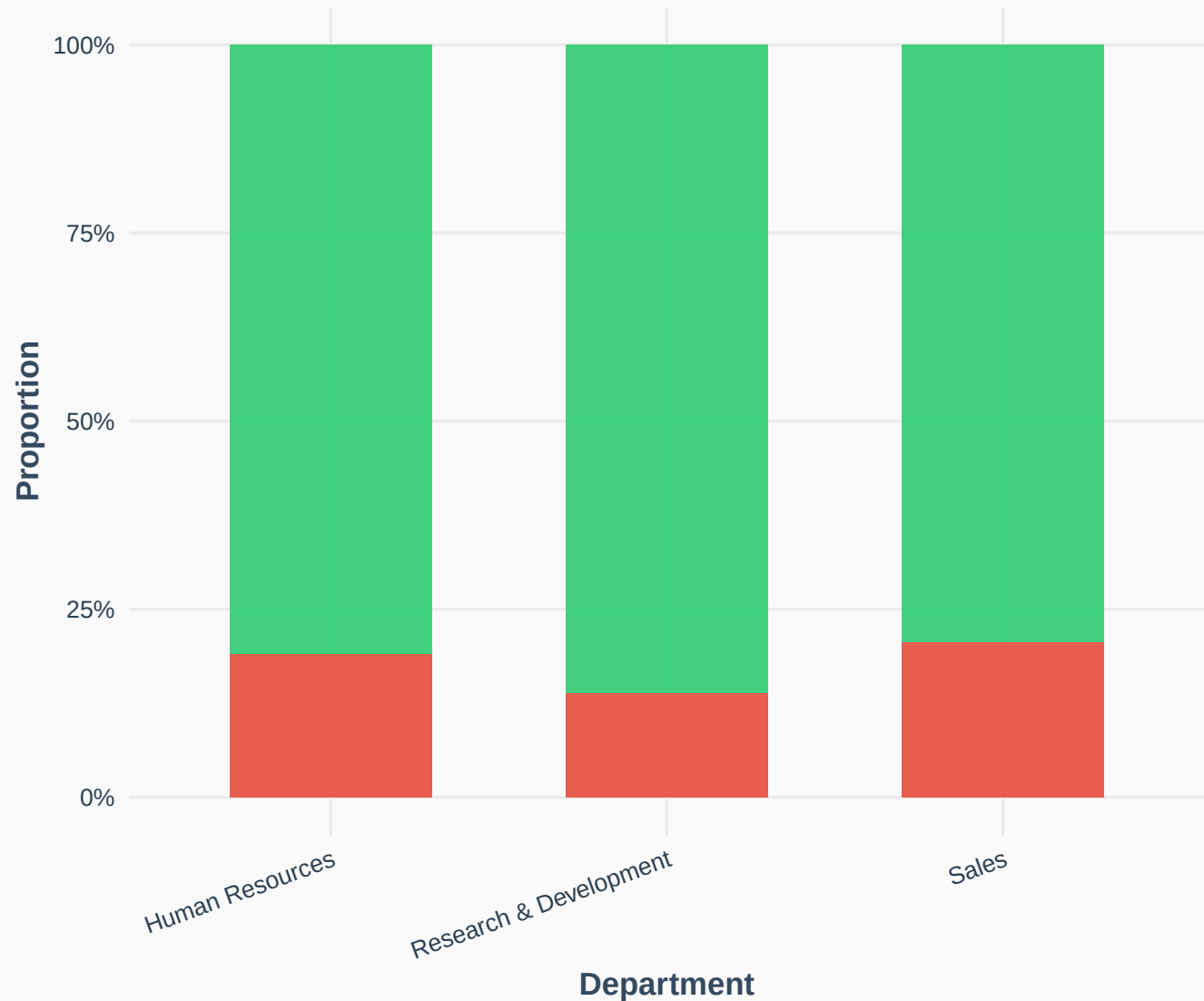
Attrition ■ No ■ Yes



Attrition Rate by Department

Comparing turnover across organizational departments

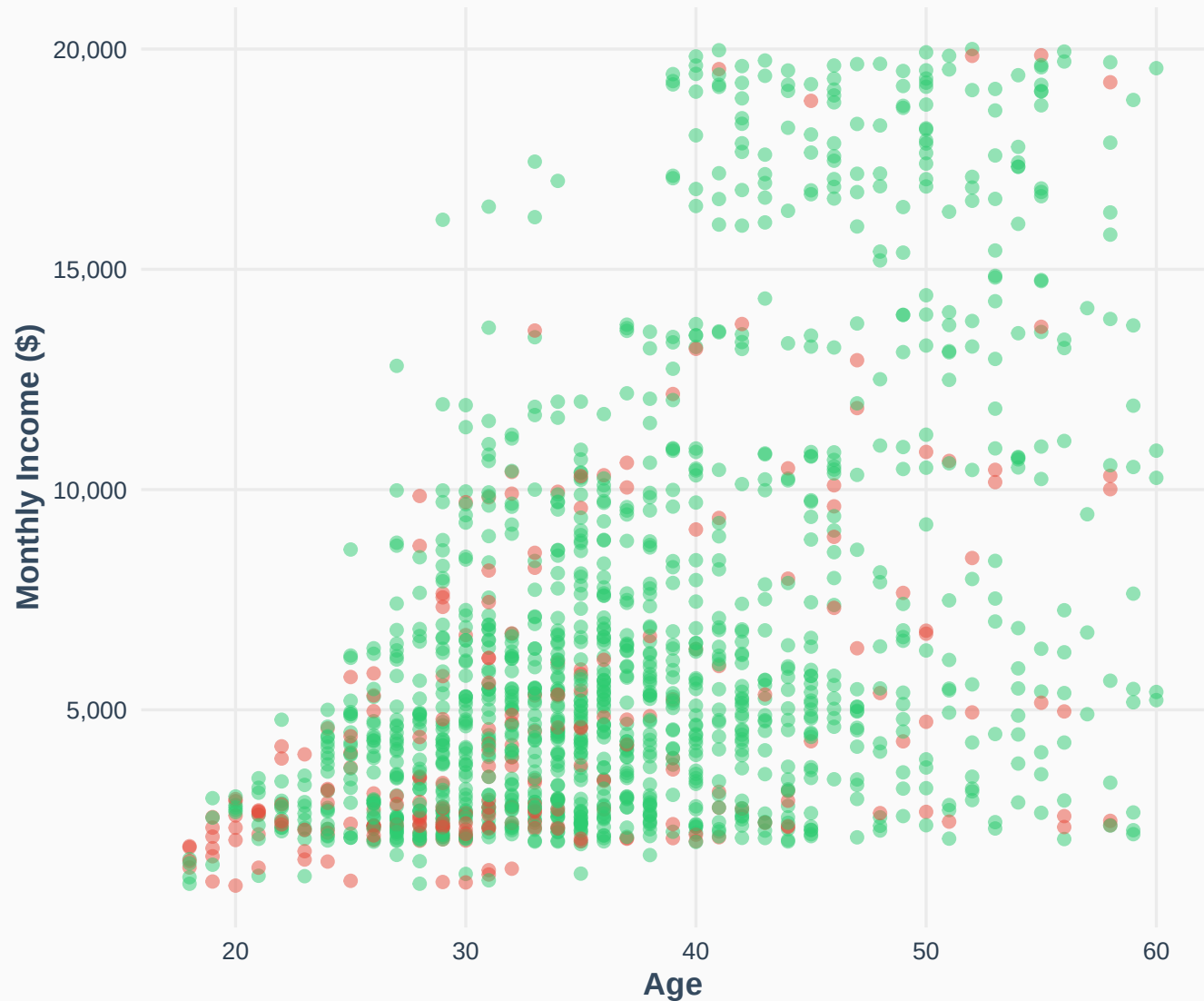
Attrition ■ No ■ Yes



Age vs. Monthly Income by Attrition

Younger, lower-income employees have higher attrition

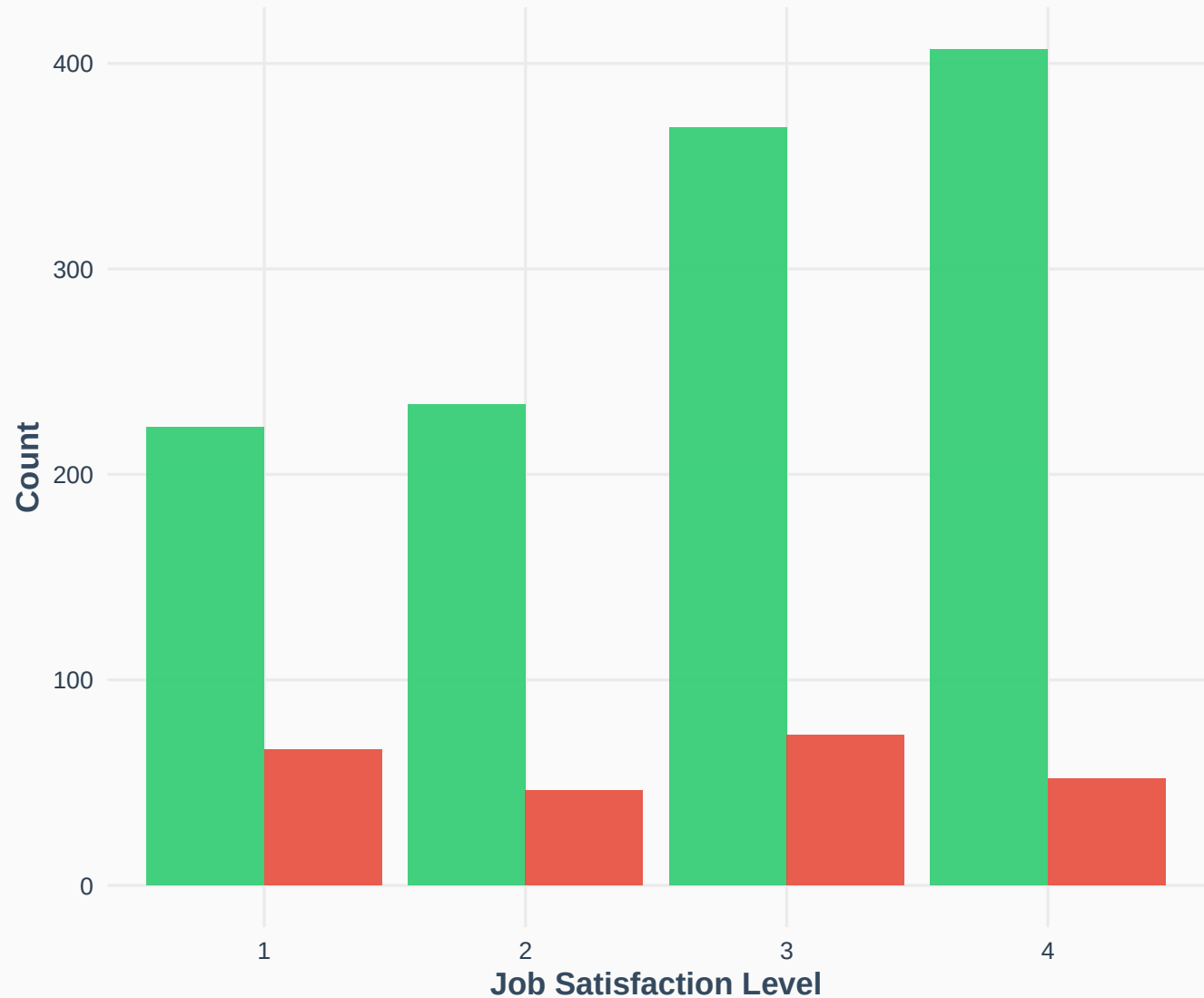
Attrition ● No ● Yes



Job Satisfaction by Attrition Status

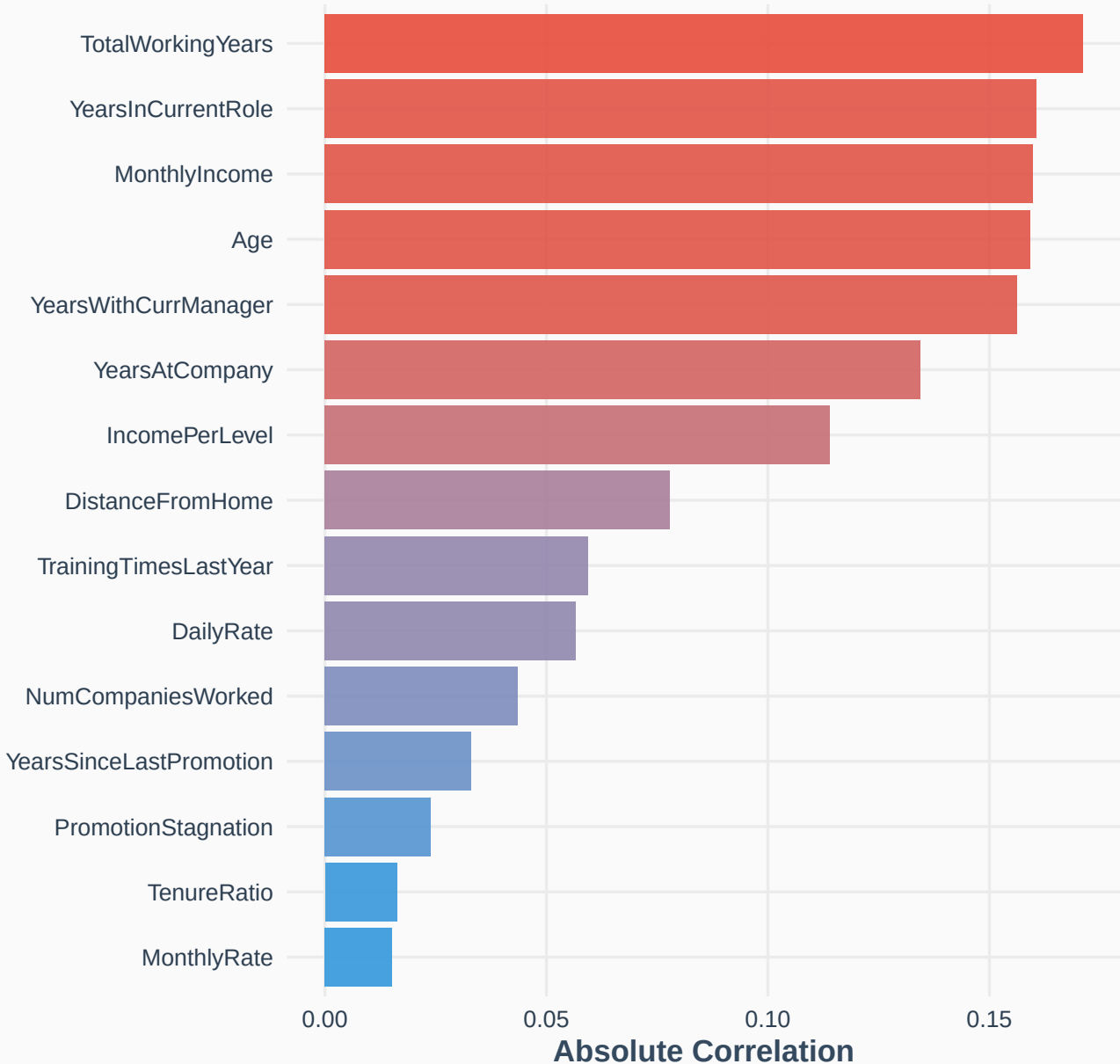
1 = Low, 2 = Medium, 3 = High, 4 = Very High

Attrition ■ No ■ Yes



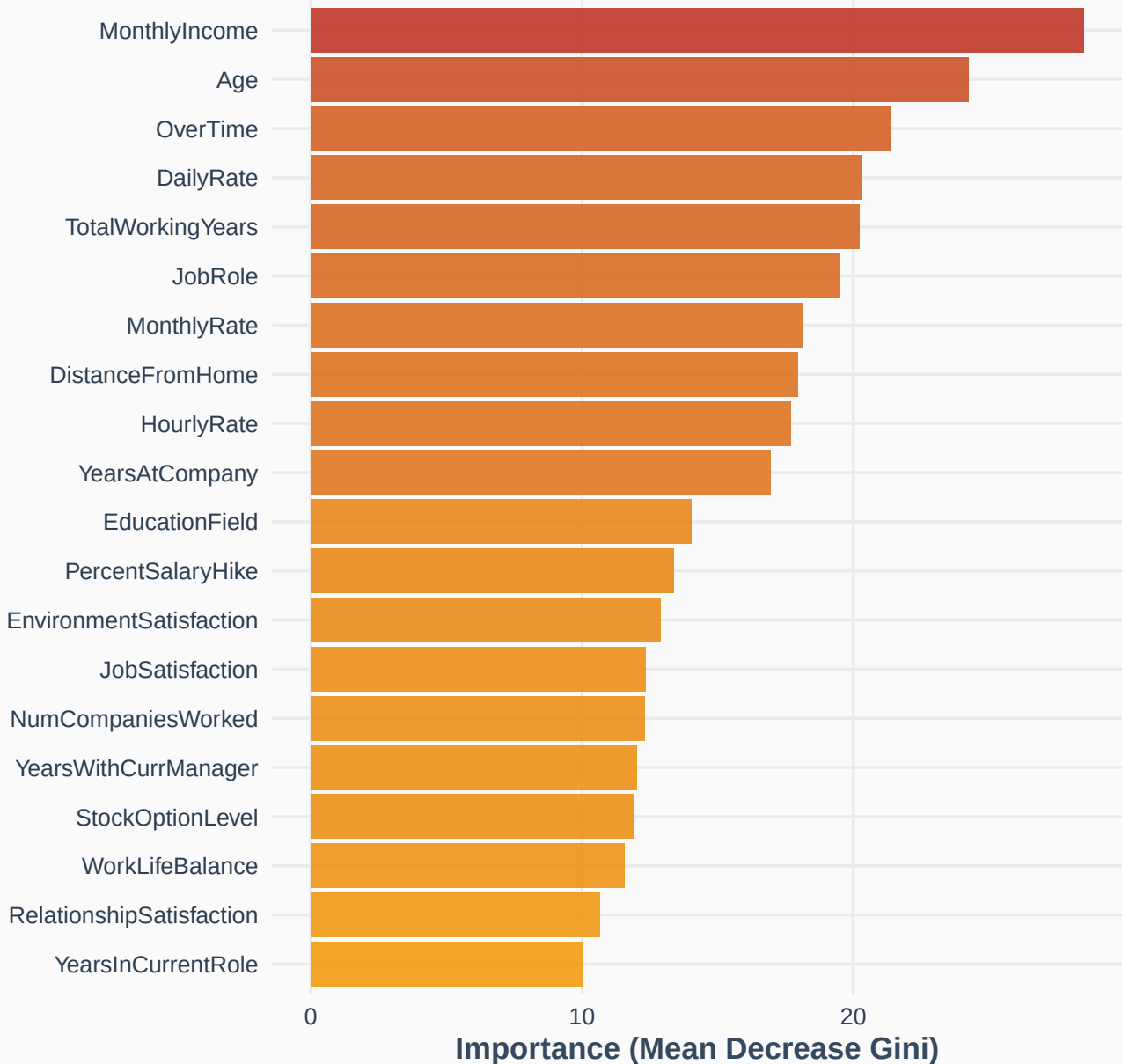
Top Features Correlated with Attrition

Absolute Pearson correlation values



Top 20 Features by Importance (Random Forest)

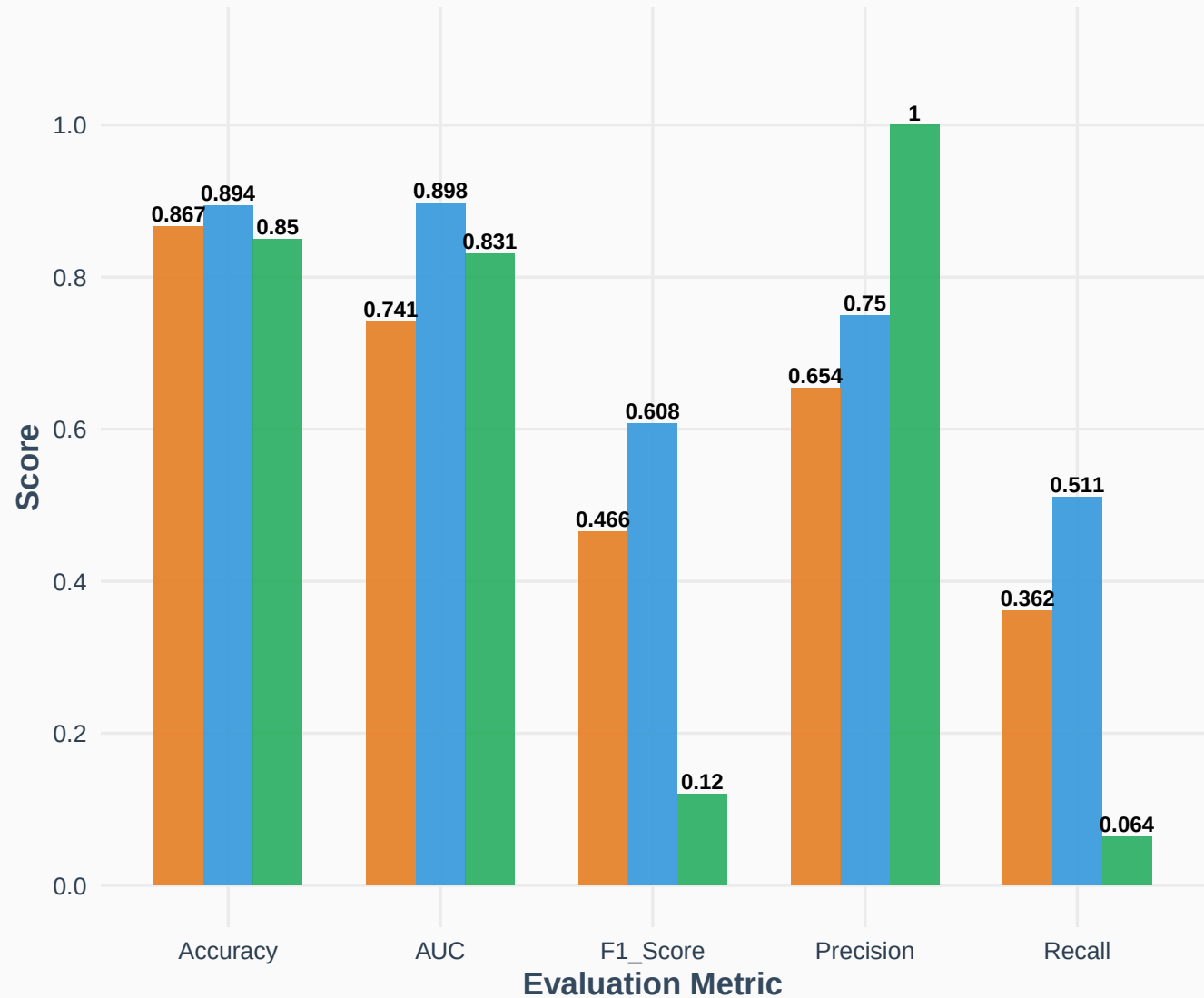
Based on Mean Decrease in Gini impurity

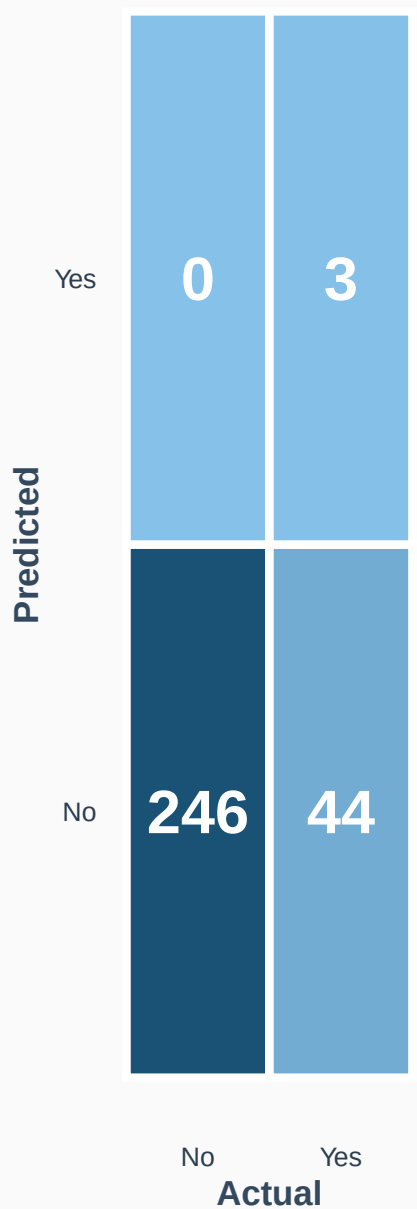
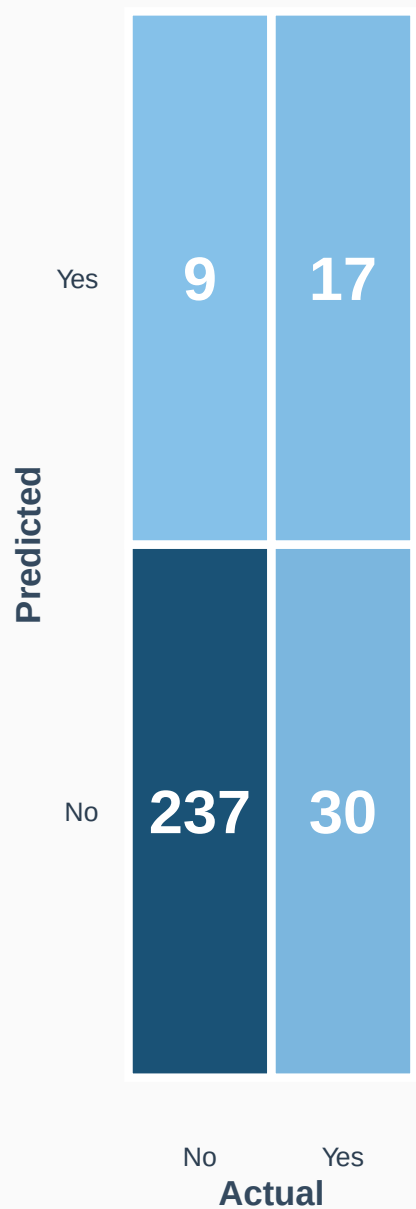
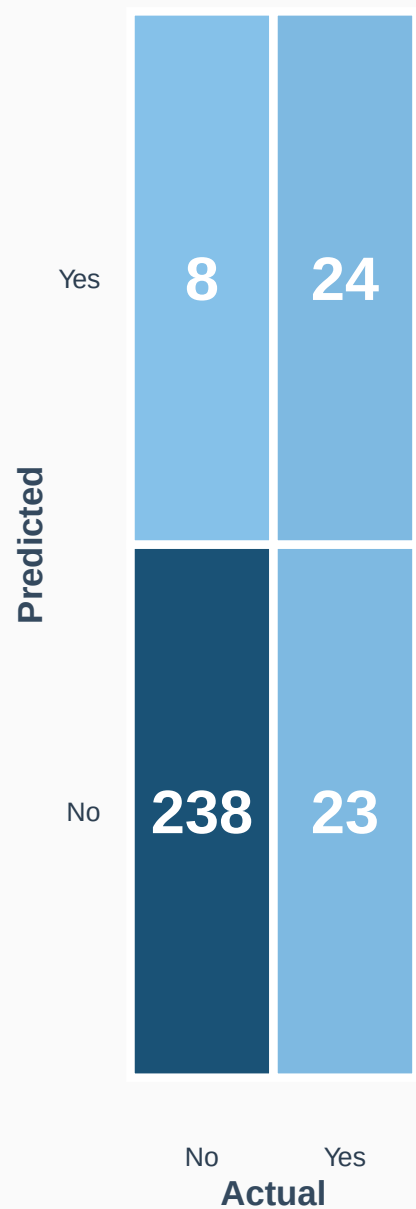


Model Performance Comparison

Evaluated on held-out test set

Model ■ Decision Tree ■ Logistic Regression ■ Random Forest





Distribution of Predicted Attrition Probabilities

Model: Logistic Regression

Attrition ■ No ■ Yes

